

4.13.61

EE25BTECH11057 - Rushil Shanmukha Srinivas

Problem: A rectangle $PQRS$ has its side PQ parallel to the line $y = mx$ and vertices P, Q and S on the lines $y = a, x = b$ and $x = -b$, respectively. Find the locus of the vertex R .

Solution:

$$y = mx \implies \mathbf{n}^\top \mathbf{x} = 0, \mathbf{n} = \begin{pmatrix} -m \\ 1 \end{pmatrix} \quad (0.1)$$

The equation of PQ is $\mathbf{n}^\top \mathbf{x} = c$

Let

$$\mathbf{P} = \begin{pmatrix} x_1 \\ a \end{pmatrix}, \mathbf{Q} = \begin{pmatrix} b \\ y_1 \end{pmatrix}, \mathbf{R} = \begin{pmatrix} X \\ Y \end{pmatrix}, \mathbf{S} = \begin{pmatrix} -b \\ y_2 \end{pmatrix} \quad (0.2)$$

In rectangle diagonals bisect each other

$$\mathbf{P} + \mathbf{R} = \mathbf{Q} + \mathbf{S} \implies \begin{pmatrix} x_1 + X \\ a + Y \end{pmatrix} = \begin{pmatrix} 0 \\ y_1 + y_2 \end{pmatrix} \quad (0.3)$$

$$x_1 = -X, a + Y = y_1 + y_2 \quad (0.4)$$

Also

$$\mathbf{n}^\top \mathbf{P} = \mathbf{n}^\top \mathbf{Q} = c \implies \begin{pmatrix} -m & 1 \end{pmatrix} \begin{pmatrix} -X \\ a \end{pmatrix} = \begin{pmatrix} -m & 1 \end{pmatrix} \begin{pmatrix} b \\ y_1 \end{pmatrix} \quad (0.5)$$

$$mX + a = -mb + y_1 \implies m(X + b) + a = a + Y - y_2 \quad (0.6)$$

$$m(X + b) + y_2 = Y \quad (0.7)$$

$$\mathbf{Q} - \mathbf{P} = \begin{pmatrix} b + X \\ y_1 - a \end{pmatrix}, \mathbf{S} - \mathbf{P} = \begin{pmatrix} X - b \\ y_2 - a \end{pmatrix} \quad (0.8)$$

In rectangle PQRS,

$$\mathbf{PQ} \perp \mathbf{PS} \implies (\mathbf{Q} - \mathbf{P})^\top (\mathbf{S} - \mathbf{P}) = 0 \quad (0.9)$$

$$\begin{pmatrix} b + X & Y - y_2 \end{pmatrix} \begin{pmatrix} X - b \\ y_2 - a \end{pmatrix} = 0 \quad (0.10)$$

$$\begin{pmatrix} b + X & m(X + b) \end{pmatrix} \begin{pmatrix} X - b \\ Y - m(X + b) - a \end{pmatrix} = 0 \quad (0.11)$$

$$(X + b)(X - b + mY - m^2(X + b) - ma) = 0 \quad (0.12)$$

$$X + b = 0 \text{ or } (1 - m^2)X + mY - (b(1 + m^2) + am) = 0 \quad (0.13)$$

$X + b = 0$ is not locus of vertex **R** as it gives a fixed point.

The locus is

$$(1 - m^2)X + mY - (b(1 + m^2) + am) = 0 \quad (0.14)$$

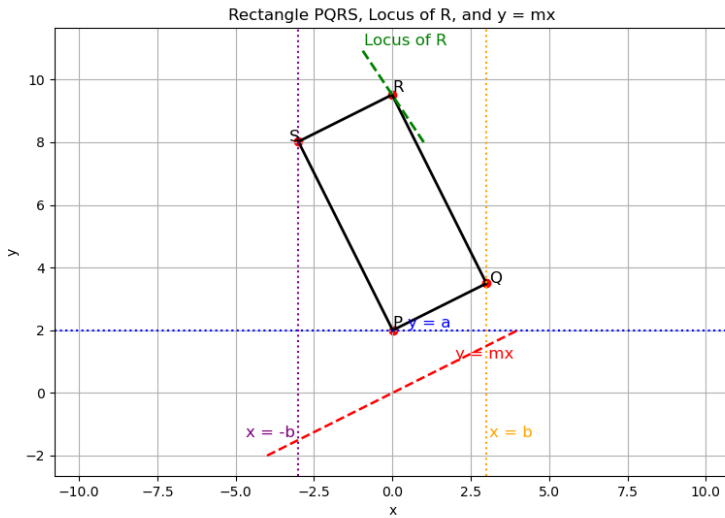


Fig: Representation of Rectangle and Locus